

# Retrofitting a commercial RF induction generator into a computer-controlled, vacuum-integrated annealing system for reactive-metal grain growth

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High-temperature vacuum annealing near a metal’s melting point drives controlled grain growth but normally requires expensive turn-key vacuum induction furnaces. We describe an open, generator-agnostic modernization stack that converts a bare commercial RF induction generator into a computer-controlled, vacuum-integrated, pyrometer-feedback annealing system. The contribution is the reproducible control/sensing/vacuum layer — LabVIEW/DAQ analog power control, dual-wavelength optical temperature feedback, a high-vacuum quartz-tube chamber, and an in-house machined graphite crucible/susceptor — which transfers between induction generators provided the generator exposes a monotonic analog power-control input. The reproducible build is anchored on the lab’s current, USA-sourced generator (CEIA “Power Cube” 6kW, procured through East Coast Induction). We report the system design, construction and operating procedures, and a performance validation grounded in the archived furnace-run data and the associated SEM/EBSD/optical characterization: a fixed-geometry power–temperature calibration ( $R^2 = 0.991$  over 1200–1400 °C), repeatable 1200 °C / 12 h Ni4N5 processing ( $1201.2 \pm 1.3$  °C across 8 runs), stable long-duration control up to 40 h at 1325 °C, and SEM/EBSD/optical microstructural characterization of annealed reactive-metal specimens cross-referenced to their logged thermal histories. A tantalum-susceptor variant of the same charge stack extends the furnace to non-coupling ceramics at 2000–2350 °C, demonstrated with YSZ grain growth and an empirical high-temperature material-compatibility map. Complete design files, a costed bill of materials, control software, and the full run-log and characterization archives are openly available.

Keywords: induction furnace; grain growth; vacuum annealing; LabVIEW; pyrometer; open-source hardware

**Status:** working draft generated from the repository’s design files, standard operating procedure, parts list, schematics, and run logs, retargeted from HardwareX to *Review of Scientific Instruments* (REVTeX 4.2, `aip,rsi`). **[TODO: ]**-marked items flag values or figures that still need to be confirmed or added before submission; outstanding bill-of-materials corrections from co-author review are tracked in the pull request and in the supplementary bill of materials.

## I. INTRODUCTION

High-temperature annealing near a metal’s melting point drives controlled grain growth, which is central to studying microstructure–property relationships in metals such as nickel and iron<sup>1–5</sup>. Doing this without oxidizing the sample requires both high temperature ( $\sim 1400$ – $1500$  °C) and high vacuum (or inert atmosphere)<sup>6,7</sup>. Commercial vacuum induction annealing furnaces that meet these requirements are expensive turn-key systems, which puts them out of reach for many academic labs.

This work takes the opposite approach: rather than building or buying a complete furnace, we retrofit and modernize a bare RF induction generator<sup>8,9</sup> into a full annealing system by adding computer power control, closed-loop optical temperature control, a high-vacuum quartz-tube chamber, and a machined graphite crucible/susceptor. The emphasis of the contribution is the modernization stack itself, which is deliberately generator-agnostic: the same control software, vacuum

chamber, pyrometer feedback, crucible, and work-coil geometry transfer from one induction generator to another. The only requirement a reader’s generator must satisfy is an analog power-control input with a monotonic command→power response over the usable range (e.g. 4–20 mA, 0–5 V, or 0–5 mA); near-linearity is convenient but not required, because the pyrometer feedback and PID calibration absorb any nonlinearity. That portability is the reproducible result and is what makes the design reusable by other labs that already own, or can buy, such an induction generator.

A second feature that broadens the system’s applicability is the machined graphite crucible/susceptor: because graphite couples strongly to the RF field and heats the charge indirectly by conduction and radiation, the furnace is not limited to electrically conductive samples — it can also process poorly- or non-coupling materials such as YSZ ceramic. This makes the same build useful well beyond reactive-metal annealing.

The modernization was first prototyped on a  $\sim 1970$  LEPEL high-frequency induction furnace owned by the BYU Mechanical Engineering department — at the time the only departmental machine able to reach the Ni/Fe melting range. A “remote control” capability was added in August 2019 to drive the generator’s power from a computer instead of the front-panel knob. The lab subsequently moved the same stack onto a current, USA-sourced solid-state generator (CEIA “Power Cube” 6kW via East Coast Induction), whose Power Controller C-V3 Plus accepts a standard analog (4–20 mA) power

setpoint. An intermediate, separately-purchased import generator (CYSI GP-15A) was evaluated but did not meet the lab’s control requirements — the required analog remote-control input was lost across vendor model substitutions — and was retired. The reproducible build documented here therefore targets the current USA generator, with the LEPEL retrofit retained as historical context.

Compared with commercial vacuum annealing furnaces, this approach has substantially lower capital cost (the generator + heating head + controller + chiller package was quoted at  $\approx$ \$16.5k in 2021, and the retrofit components are itemized in the supplementary bill of materials, versus \$50k–\$200k+ for turn-key systems), is repairable and modifiable by the user, and is broadly reusable: any lab with an induction generator and an analog power input can reproduce the control, vacuum, and temperature-sensing additions described below. The system is aimed at materials labs needing oxidation-free grain-growth anneals of reactive metals (Ni, Fe) near their melting points; at labs that own a bare induction generator and want to add computer setpoint control, optical feedback, and vacuum integration without buying a turn-key furnace; at groups processing poorly- or non-coupling charges (e.g. YSZ ceramic) that can be heated indirectly through the graphite susceptor; and at teaching/maker contexts wanting a repairable, modifiable high-temperature vacuum furnace built from documented, open design files.

All design files, control software, run logs, and characterization data are openly available: the development repository is <https://github.com/vertical-cloud-lab/custom-induction-furnace> and an archival snapshot is deposited at Zenodo ([10.5281/zenodo.20878017](https://doi.org/10.5281/zenodo.20878017)), with hardware design files under the CERN-OHL-S v2 license and control code under the MIT license (proposed).

### A. Scope and reproducibility boundary

To be explicit about what this paper does and does not reproduce: this work does *not* reproduce the RF generator internals. It reproduces the modernization layer built around a compatible generator — the computer power control, optical (pyrometer) feedback, high-vacuum quartz-tube chamber, machined graphite crucible/susceptor, work-coil geometry, and the mechanical support fixturing. A reader supplies (or already owns) a compatible generator and rebuilds the layer described here.

### B. Prior art and comparison

Three classes of alternative exist, and each presents a limitation this design addresses. Commercial vacuum induction annealing furnaces are turn-key systems

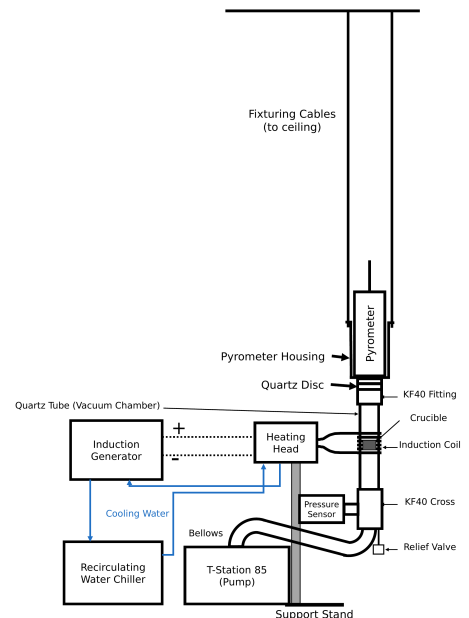


FIG. 1. System overview of the computer-controlled, vacuum-integrated induction annealing system. The induction generator drives the water-cooled heating head/work coil, mounted on its support stand, around the vertically suspended quartz-tube vacuum chamber holding the crucible; cooling water flows from the recirculating chiller through the heating head and then the generator in series before returning to the chiller. The chamber stack is joined by KF40 flanges to the pyrometer housing, which hangs from the ceiling by fixturing cables, and the pyrometer sights the charge from above through a quartz optical disc. At the bottom of the stack, an overpressure relief valve and a flexible bellows lead past the pressure sensor to the T-Station 85 turbo pumping station. Generator power is commanded, and the pyrometer read back, by the LabVIEW/DAQ control path described in the text.

costing \$50k–\$200k+; they are closed, difficult to modify or repair in-house, and carry vendor lock-in. Openly documented resistively heated tube and box furnaces reach lower maximum temperatures with slower ramps than induction heating, and offer no RF-susceptor route to heating a non-coupling charge. Finally, induction generators themselves are widely available as bare industrial units (CEIA, Ambrell, and others), but ship without integrated vacuum, optical temperature feedback, or computer setpoint control — exactly the layer this work adds.

## II. SYSTEM DESIGN AND DESCRIPTION

### A. Architecture and control signal flow

The system comprises five subsystems, shown in the labeled overview schematic of Fig. 1; Fig. 5 shows

the corresponding assembled system, with its main components labeled, at operating power.

The heating subsystem is a solid-state induction generator (current build: CEIA Power Cube PW3-90/50) driving a water-cooled copper work coil that carries both the oscillating current and the cooling water. The coil geometry ( $\sim 3$  in tall, 2.5 in ID,  $\sim 6.5$  turns) was tuned to the sample/crucible stack<sup>10,11</sup>.

The vacuum and gas subsystem starts from the quartz tube that forms the chamber, connected through a KF40 flange and flexible bellows to an Edwards T-Station 85 turbo-molecular pump ( $\sim 1 \times 10^{-6}$ – $10^{-8}$  Torr). A manual vent valve fed from an inert-gas regulator — optionally through a mass flow controller, as in this build — allows controlled venting and backfilling to suppress oxidation of reactive metals<sup>12,13</sup>. The choice of atmosphere matters for graphite-crucible anneals: the best results were achieved by pumping the chamber to high vacuum and then backfilling with argon through the vent valve — purging the chamber with the inert gas before settling to a steady trickle flow ( $\sim 20$  SCCM, metered here by the optional mass flow controller) for the anneal. Heating under vacuum alone instead tended to sublime graphite from the crucible onto the chamber walls. For the highest-temperature work, such as the temperatures needed for ceramic grain growth (Sec. V A), the chamber was evacuated and then continuously filled with argon while the turbopump was running, which helped avoid oxidation during heating.

Temperature is sensed optically: a dual-wavelength (ratio) pyrometer<sup>14,15</sup> (the canonical unit is a LumaSense IMPAC ISR 6, 800–2500 °C; the IMPAC brand is LumaSense’s) views the sample through a KF40 optical window, providing a non-contact temperature signal that is robust to emissivity changes and to the RF field that would corrupt thermocouples<sup>16–18</sup>.

Control is implemented in LabVIEW: a DAQ analog output sets generator power, and the pyrometer signal is read back for closed-loop ramp/soak control, with VIs for manual control, automated ramping, PID tuning, and email alerts. The front panel exposes editable ramp-up, PID-soak, and cool-down setpoint tables, the PID gains, the inert-gas flow setpoint/readback (when the optional mass flow controller is installed), and live power-command and pyrometer-temperature traces used to run and monitor an anneal. In the CEIA build the DAQ outputs 0–5 V to a voltage-to-current loop conditioner that converts the command to the controller’s 4–20 mA power-setpoint input (the LEPEL prototype instead took a 0–5 mA command directly), so the closed-loop signal flow runs from the LabVIEW setpoint through the DAQ analog output and V→I conditioner to the generator power, and back from the sample/crucible heating through the pyrometer optical reading and DAQ analog input (scaled to °C) into the PID controller, which updates the analog command. The same path runs open-loop (direct command ramp) while the temperature is below the pyrometer’s  $\sim 700$  °C detection floor; once the

pyrometer reports a sustained valid reading above that floor, control hands over to closed-loop PID on the optical temperature<sup>19,20</sup>.

Mechanically, a bolted support stand (8 short + 4 long sections, 12 corner braces, 4 floor mounts) carries the generator’s heating head, centering the work coil on the crucible position. The vacuum chamber itself has no support stand: the chamber stack is joined by KF40 flanges to the pyrometer housing, which is suspended from the ceiling by cables, so the vertical stack hangs from above. An in-house machined graphite crucible holds the sample and acts as the RF susceptor.

## B. Technical specifications

As built, the documented system heats charges inside a 35 mm-ID fused-quartz tube at an operating vacuum level of  $\sim 1 \times 10^{-6}$  to  $1 \times 10^{-8}$  Torr, with temperature measured by the dual-wavelength ratio pyrometer over its  $\sim 800$ – $2500$  °C detection range (no reading below  $\sim 700$  °C). Directly coupled metal charges in the machined graphite crucible reach the Ni/Fe grain-growth range ( $\sim 1400$ – $1500$  °C), with demonstrated soaks from minutes up to  $\sim 40$  h in the archived run logs; exchanging the graphite stack for the tantalum-susceptor ceramic configuration extends the working range to  $\sim 2000$ – $2350$  °C (Sec. V A). Power is commanded through a 4–20 mA analog setpoint via the LabVIEW/DAQ output and V→I conditioner (the LEPEL prototype instead took 0–5 mA directly), and the water-cooled copper work coil measures  $\sim 3$  in tall, 2.5 in ID, with  $\sim 6.5$  turns. These are *build-dependent* values: a reader supplies their own generator, pump, and pyrometer and re-establishes the corresponding values for their build.

## C. Generator compatibility and portability

The control/sensing/vacuum layer transfers to any generator that exposes an external analog power-control input (e.g. 4–20 mA or 0–5 V/0–5 mA) whose command→delivered-power response is monotonic over the usable range (near-linear is convenient but not required, since the pyrometer feedback closes the loop); that supports a water-cooled induction head/coil matching the documented coil dimensions; that delivers enough RF power to reach the target temperature for the documented crucible/sample stack; that allows safe electrical isolation between the DAQ/control path and the generator input; and for which the required cooling and line power are available.

When porting to a new generator, the following must be re-established: analog-command scaling, power-to-temperature calibration, PID gains, coil/sample alignment, maximum safe ramp rate, and any generator-specific interlock behavior. The port from the  $\sim 1970$  LEPEL prototype to the current CEIA build

is the existence proof for this portability claim: the dual-wavelength pyrometer feedback, the quartz-tube/KF40 vacuum chamber, the machined graphite crucible/susceptor, the bolted support stand, and the LabVIEW/DAQ control VIs all transferred unchanged. Only the analog interface itself differed — the LEPEL took a 0–5 mA command directly, whereas the CEIA Power Controller C-V3 Plus expects 4–20 mA, hence the V→I loop conditioner — and the power–temperature calibration and PID gains were re-established for the new generator, as they must be for any port. The generator’s internal operating frequency ( $\sim 200$ – $450$  kHz vacuum-tube LEPEL vs. the solid-state CEIA’s self-set frequency) and maximum power ( $\sim 35$  kW vs. 6 kW) sit behind the analog interface and required no change to the stack beyond re-tuning power for the charge.

#### D. Custom-machined graphite crucible / susceptor

Because some charges of interest couple poorly to the RF field (e.g. YSZ ceramic) or must be physically contained, the sample is held in a graphite crucible machined in-house from  $3000^\circ\text{C}$ -grade graphite stock (Cotronics 56L-3). The graphite couples strongly to the induction field and acts as a susceptor, transferring heat to the contained charge by conduction/radiation; this makes the same furnace usable for conductive metals (direct coupling) and for non/poorly-coupling materials (indirect, susceptor-driven heating). Crucibles are machined to fit inside the quartz tube and seat on a ceramic-cylinder / Teflon spacer stack. Cracked crucibles are repaired with a  $3000^\circ\text{C}$  graphite cement (Resbond 931) rather than discarded. The two-piece geometry — a cup body and a lidded cap with a central bore — contains the charge while leaving an optical path for the pyrometer to sight the sample (Fig. 2). The measured dimensions of every piece in the stack (crucible body, lid, alumina spacer discs, and sapphire pyrometer window) are given in Fig. 3; fully assembled, the crucible stack stands 13 mm tall. The specimen-loading sequence — specimen sandwiched between two alumina sheets in the cup body, lid seated on top, and sapphire window placed over the lid bore — is shown step by step in Fig. 4. For ceramic charges requiring soak temperatures beyond the graphite/alumina stack’s service range, the crucible stack is exchanged for a tantalum-susceptor variant described in Sec. VA.

### III. CONSTRUCTION AND OPERATION

#### A. System assembly

This section describes how the subsystems of Fig. 1 are physically connected, following the heating, vacuum, gas, and control paths in turn; the costed part list and the design-file inventory (coil drawing,



FIG. 2. Fully disassembled graphite crucible. Top left is the crucible lid, top middle is the bottom of the crucible that acts as the specimen carrier, far right is the sapphire window, bottom left and middle are the alumina sample-surrounding sheets.

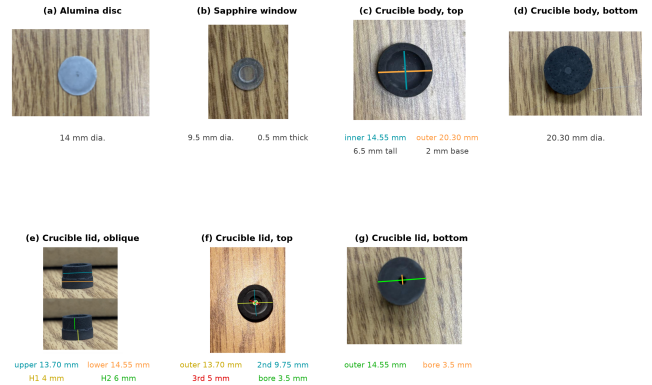


FIG. 3. Measured dimensions of each part of the graphite crucible/susceptor stack, with the coloured measurement lines drawn on the corresponding photograph. (a) Alumina spacer disc, 14 mm dia. (b) Sapphire pyrometer window, 9.5 mm dia., 0.5 mm thick. (c) Crucible body, top: machined sample cavity 14.55 mm ID (teal), 20.30 mm OD (orange); the cup is 6.5 mm tall with a 2 mm base. (d) Crucible body, bottom face, 20.30 mm dia. (e) Crucible lid, oblique: the stepped plug seats into the cavity — 13.70 mm upper step (teal), 14.55 mm lower shoulder (orange); the upper step is 4 mm tall (H1, yellow) and the lower shoulder 6 mm tall (H2, green). (f) Crucible lid, top: four concentric steps of 13.70 mm (outermost), 9.75 mm, 5 mm, and a 3.5 mm central pyrometer-sighting bore. (g) Crucible lid, bottom: 14.55 mm seating face (green) and the 3.5 mm through-bore (orange). Two alumina discs sandwich the specimen inside the cavity, the lid seats on top, and the sapphire window closes the bore so the pyrometer sights the sample through the lid; fully assembled the stack stands 13 mm tall.

editable schematics, STEP models) are maintained as supplementary material. Figure 5 shows the assembled system at operating power, and Fig. 7 details the vacuum and gas-handling connections described below.

*a. Heating path.* The generator’s heating head, carrying the water-cooled copper work coil ( $\sim 3$  in tall,



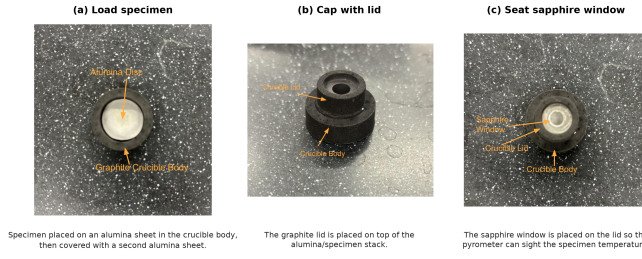


FIG. 4. Specimen-loading sequence for the graphite crucible/susceptor, in order (orange call-outs name the visible parts). (a) The specimen is placed on an alumina sheet in the crucible body and covered with a second alumina sheet, so the alumina discs isolate the specimen from the graphite. (b) The graphite lid is seated on top of the alumina/specimen stack. (c) The sapphire window is placed over the lid bore, closing the crucible while leaving an optical path for the pyrometer to sight the specimen temperature through the window.

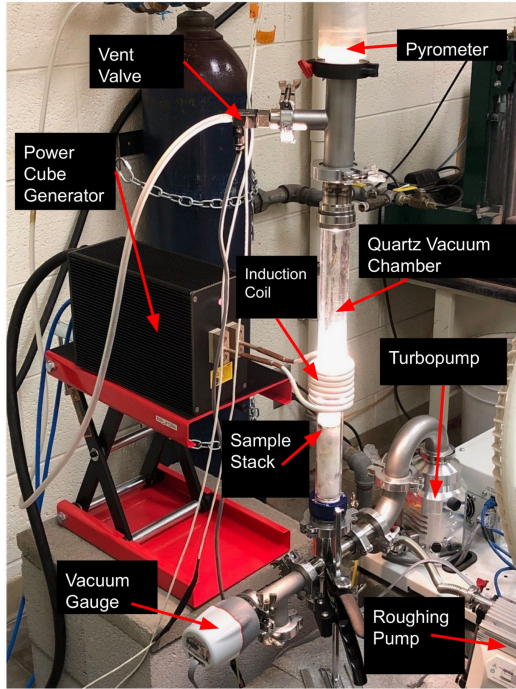


FIG. 5. The fully assembled, functioning furnace with the main components labeled. The Power Cube generator (left) drives the water-cooled induction coil around the vertical quartz vacuum chamber, which glows at the sample-stack position; the pyrometer sights down the chamber axis from the top of the stack; the vent valve admits the argon backfill; and the wide-range vacuum gauge, turbopump, and separately supported roughing pump complete the vacuum path at the bottom of the stack.

2.5 in ID,  $\sim 6.5$  turns; Fig. 6), mounts on an adjustable platform on the bolted support stand (8 short + 4 long sections, 12 corner braces, 4 floor mounts) so the coil can be centered on the crucible position of the vertically suspended quartz tube. Cooling water from the recirculating chiller flows first through the heating

head and work coil, then through the generator, before returning to the chiller; the coil conductors carry both the RF current and the coolant. Inside the tube, the machined graphite crucible/susceptor (Figs. 2–4), machined from 3000 °C graphite stock and repaired with graphite cement when cracked, seats at coil height on a ceramic-cylinder/Teflon spacer stack; the pyrometer, mounted in its housing at the top of the chamber stack, sights the specimen through the optical window and the sapphire disc in the crucible lid.

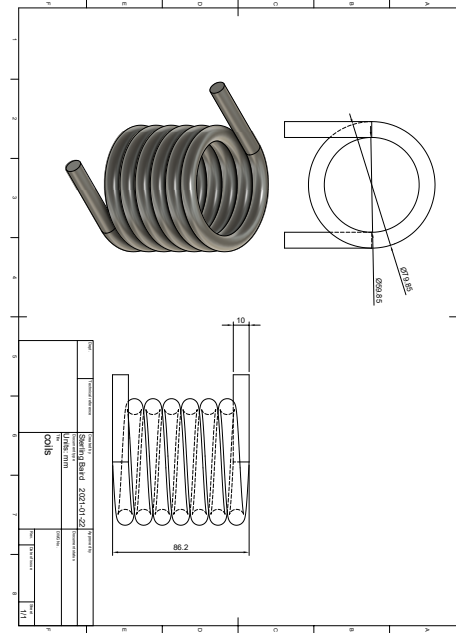


FIG. 6. Work-coil fabrication drawing used for the CEIA build, showing the coil geometry and dimensional callouts used to form and mount the water-cooled copper coil around the quartz chamber.

*b. Vacuum path.* The chamber is the fused-quartz tube itself. Its open end terminates in a KF40 flange whose optical window — a 55 mm quartz disc seated on the overpressure centering ring and held by a reusable plastic quick-clamp (a Torr-Seal'd blank flange is the documented permanent-seal alternative) — closes the top of the stack. Every flanged joint in the chamber stack is sealed the same way: the mating KF40 faces compress an elastomer O-ring carried on a centering ring, so the stack assembles and opens with hand-tightened clamps and no tools, and a worn seal is renewed by swapping a single ring. These elastomer O-ring seals support the system's  $\sim 1 \times 10^{-6}$ – $10^{-8}$  Torr operating vacuum; all-metal (CF) sealing would be required only for ultra-high-vacuum service beyond what the annealing application needs. The chamber stack rests on no support stand: its components are connected by KF40 flanges to the pyrometer housing, which is suspended from the ceiling by cables, so the whole vertical stack hangs from above. From the bottom of the chamber stack, a tee carries a

0.5 psi overpressure relief valve (Fig. 7(b)) that protects the quartz tube from accidental overpressure during backfilling, and a flexible stainless bellows continues through the wide-range vacuum gauge to the Edwards T-Station 85 turbo pumping station. The roughing pump sits on a separate support, mechanically decoupled from the vacuum-chamber attachment tubes, so its vibration does not reach the chamber and the pyrometer sight line (Fig. 7(a)).

*c. Gas path.* The argon cylinder regulator feeds the manual vent valve, providing both controlled venting and low-flow inert backfilling of the chamber. A digital mass flow controller between the regulator and the vent valve is optional but convenient: in this build a Sierra Smart-Trak meters the backfill trickle, and its flow setpoint and readback are exposed on the LabVIEW front panel. A regulator and needle valve can serve the same role in a minimal build.

*d. Control path.* The DAQ's 0–5 V analog output drives the voltage-to-current loop conditioner, whose 4–20 mA output lands on the CEIA Power Controller's analog power-setpoint input (the LEPEL prototype took a 0–5 mA command directly); the pyrometer's analog output returns to a DAQ analog input, scaled to °C. Electrical isolation is maintained between the DAQ/control path and the generator interface throughout. The LabVIEW VIs are deployed against these DAQ channels, and manual control is verified before automated ramp/soak is enabled.

## B. Operating procedure

The operating procedure below is condensed from the lab standard operating procedure. Before RF power is ever applied, the chamber must be at high vacuum ( $\sim 1 \times 10^{-6}$  Torr or better) or backfilled to the intended inert atmosphere; cooling-water flow at the coil and generator must be confirmed and leak-free; the coil, sample, and pyrometer sight line must be aligned; and all conductive objects (jewelry, watches, phones, tools) must be at least 6–12 in from the coil.

A run begins with the chamber vented to atmosphere — the vent valve is never opened while the turbo pump is spinning. The specimen is loaded into the graphite crucible (Fig. 4), the crucible/spacer stack is inserted into the quartz tube, and the KF40 flange is clamped finger-tight on its O-ring. With the vent valve closed, the roughing and turbo pumps are brought up and the chamber pumped to high vacuum. The generator is then powered on per its own startup sequence and the LabVIEW ramp/soak profile started. Below the pyrometer's  $\sim 700$  °C detection floor the VI ramps open-loop on the analog command; once the pyrometer returns a sustained valid reading above that floor, control hands over to closed-loop PID on the optical temperature, and the programmed profile (e.g.,  $\sim 1200$  °C for several hours for Ni grain growth) runs to completion, with optional

email status alerts from the VI. At the end of the profile, power ramps down and RF stops; the sample cools under vacuum, and the chamber is vented only after the turbo pump has fully stopped.

Off-normal conditions all call for removing RF power first. On loss of cooling water or a chiller alarm, the analog command is zeroed immediately and the generator not re-energized until flow is restored. On vacuum loss or a quartz-tube fracture, RF is stopped immediately and the chamber is never vented against a spinning turbo pump. On arcing, smoke, or runaway temperature, the command is zeroed, RF is killed at the generator, and the system is locked out before inspection.

## C. Safety

The system combines RF power, high vacuum, high temperature, and cooling water, and each pairing carries its own hazard. The energized coil can shock or burn through direct contact or through currents induced in nearby conductors, so conductive objects and electronics are kept at least 6–12 in away, the DAQ/control path is electrically isolated from the generator interface, and any service inside the generator follows lockout/tagout. The evacuated quartz tube poses a fracture/implosion risk — operators wear safety glasses — and venting against a spinning turbo pump can destroy the pump, so the turbo must spin down fully before the vent valve opens. Because the coil and chiller run water immediately next to energized hardware, flow is confirmed and leaks are checked before RF is applied, and water lines are routed away from electronics. A melted-through specimen or hot graphite oxidizing on a premature vent can contaminate the pump line and burn the operator, so samples cool under vacuum and vacuum components are handled with clean nitrile gloves. Inert-gas backfilling in an enclosed space is an asphyxiation risk and is done only with adequate ventilation. Finally, the RF near-field can corrupt or damage nearby instruments, which are shielded and kept clear of the coil.

## IV. PERFORMANCE VALIDATION

The archive contains more than one hundred logged runs across Ni200, Ni4N5, YSZ, and Pd thermal-evaporation experiments, with temperatures of  $\sim 900$ – $1400$  °C and soak durations from minutes to  $\sim 40$  h, each with raw spreadsheet/text/LabVIEW-measurement traces and some photos. Per-run traces were parsed into uniform CSV/PNG files, and the quantitative metrics below were computed reproducibly from those traces by scripts archived alongside the data (see Data Availability). Individual runs and specimens are identified at the run-ID/specimen-ID level only in the supplementary specimen-run cross-reference, which links every entry used below to its raw log and image data

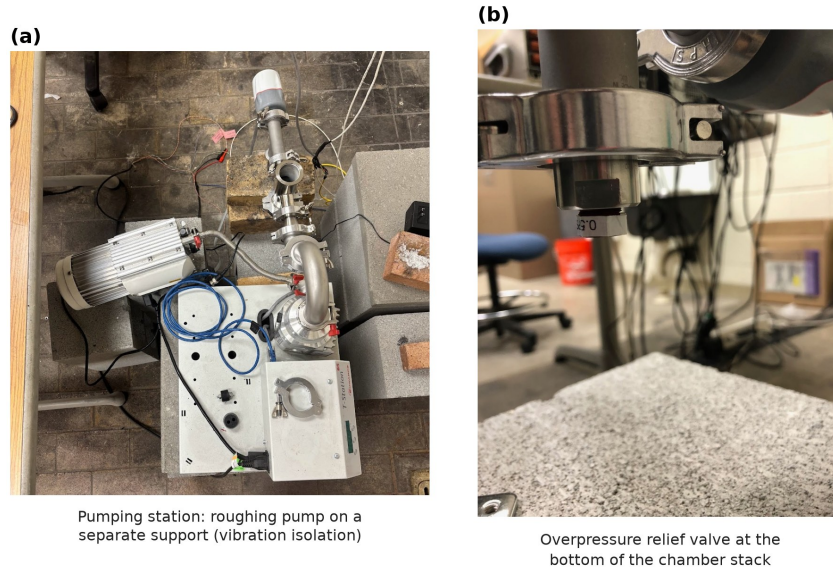


FIG. 7. Vacuum and gas-handling details of the assembled system. (a) Turbo pumping station with the roughing pump mounted on a separate support, coupled to the pumping station only through a flexible hose, so pump vibration is not transmitted into the chamber. (b) Overpressure relief valve (0.5 psi cracking pressure) at the bottom of the vacuum chamber stack.

in the archive. Because the archive is a retrospective operating record rather than a prospectively balanced design of experiments, every comparison is restricted to an internally consistent subset (matching material, nominal profile, and operating configuration) and the power–temperature relation is reported as an *operational* calibration for one fixed geometry, not as a universal transfer function.

*a. Power-command  $\rightarrow$  temperature calibration (Fig. 8).* For one fixed Ni4N5/quartz/graphite/optical configuration, four archived steady-state anneals spanning 1200–1400 °C give a monotonic relation between the soak-mean analog power command and the soak-mean pyrometer temperature, well described by  $T \approx 931 + 632 I$  (°C,  $I$  in mA),  $R^2 = 0.991$ ,  $n = 4$ . This is presented as an operational calibration only: the broader archive shows configuration-dependent offsets (e.g. runs near 1200 °C required soak-mean commands of  $\sim 0.44$  mA vs.  $\sim 0.66$  mA in a different era/geometry), so the fit must not be read as a universal command–temperature law.

*b. Representative ramp/soak with control metrics (Fig. 9).* A Ni4N5 anneal at a nominal 1300 °C / 12 h condition serves as the representative closed-loop run. Measured against the 1300 °C setpoint, the soak held  $1302.1 \pm 3.0$  °C (mean  $\pm$  SD) over the 12 h plateau, with a 10–90 % rise time of  $\sim 1135$  s, an overshoot of  $\sim 12$  °C, a maximum in-soak deviation of  $\sim 10$  °C, and a near-zero linear drift of  $-0.02$  °C/h, demonstrating that the pyrometer-feedback loop holds a programmed soak over many hours.

*c. Repeatability and long-soak stability (Figs. 10–11).* Run-to-run repeatability was assessed on the most internally consistent cohort available: eight Ni4N5

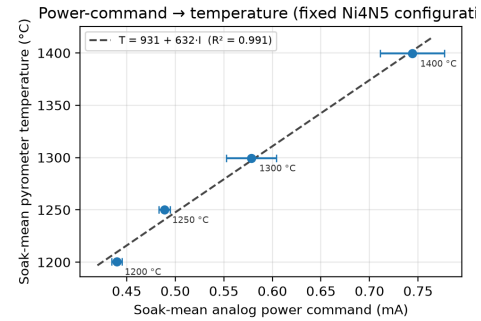


FIG. 8. Power-command  $\rightarrow$  temperature calibration for one fixed archived Ni4N5 configuration. Points are soak-mean values; horizontal bars are the in-soak SD of the analog command. The dashed line is the linear operational calibration ( $R^2 = 0.991$ ,  $n = 4$ ).

anneals at a nominal 1200 °C / 12 h condition, all logged under the same gas-flow configuration. Across these runs the soak temperature was  $1201.2 \pm 1.3$  °C (range 1199.8–1204.2 °C; coefficient of variation 0.11 %;  $n = 8$ ), peak temperature  $1212.6 \pm 6.7$  °C, and soak duration  $766 \pm 15$  min. Runs from other periods or geometries are deliberately *not* pooled into this estimate. Long-duration stability was demonstrated with two Ni200 anneals at 1325 °C: a 20 h run held  $1326.5 \pm 1.5$  °C with drift  $-0.004$  °C/h, and a 40 h run maintained  $\sim 1326$ – $1329$  °C over 2400 min in-band.



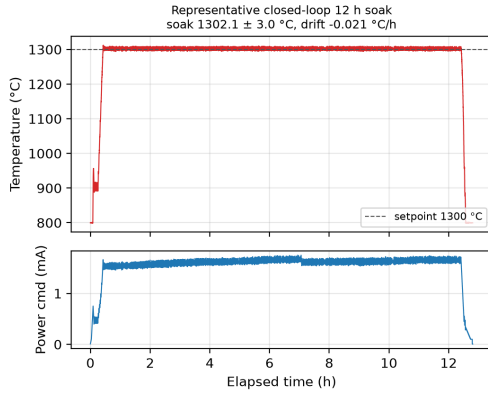


FIG. 9. Representative closed-loop anneal (Ni4N5, nominal 1300 °C / 12 h): pyrometer temperature with setpoint overlay (top) and analog power command (bottom). Soak mean  $1302.1 \pm 3.0$  °C, drift  $-0.02$  °C/h.

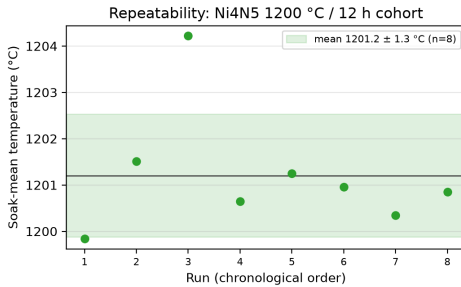


FIG. 10. Run-to-run repeatability of the Ni4N5 1200 °C / 12 h cohort ( $n = 8$ ). Points are per-run soak-mean temperatures; the band is mean  $\pm$  SD ( $1201.2 \pm 1.3$  °C).

## V. MICROSTRUCTURAL VALIDATION

The furnace exists to change microstructure, so the key validation is real microstructural data tied to specific thermal histories. The archived SEM/EBSD and optical-microscopy datasets are keyed by Ni4N5-### / Ni200.### specimen IDs that map directly to the run IDs of the anneal logs. This mapping is built reproducibly by a script that parses the specimen ID out of every characterization folder, joins it to the machine-parsed run summary, and writes the per-specimen cross-reference distributed with the supplementary material: 35 characterized specimens, of which 9 link to a parsed run log, 10 carry archived EBSD/OIM data, and one carries EDS. Inverse-pole-figure (IPF) EBSD orientation maps of two annealed specimens (Fig. 12) resolve hundreds of indexed grains. Notably, the anneals leave surfaces that index well with no metallographic preparation. Fig. 13(a,b) shows a raw, as-collected Kikuchi pattern (electron backscatter pattern) recorded live from specimen Ni4N5\_028, with bands and zone axes crisply resolved even at coarse  $8 \times 8$  detector binning, alongside the SEM live view of the thermally grooved

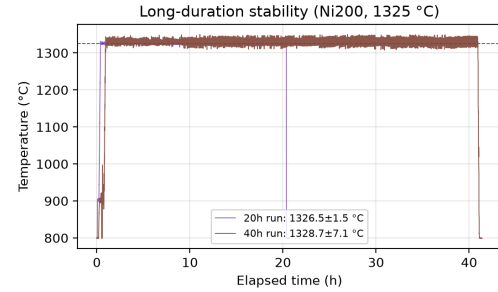


FIG. 11. Long-duration thermal stability: Ni200 anneals at 1325 °C held for 20 h and 40 h.

grain boundaries that the accompanying EBSD line scans traverse. Nor were such patterns only viewed live: during EBSD mapping of two nickel specimens from the early commissioning campaign, the acquisition software archived the raw pattern at every scan point, and more than 100,000 per-point raw patterns survive (Fig. 13(c-e)), enabling re-indexing or pattern-quality re-analysis with modern indexing approaches without re-mounting the specimens. The thermal grooving that delineates the boundaries is itself a record of the anneal: edge-on views of specimen Ni4N5\_053 (annealed at 1200 °C / 12 h), imaged after only an ultrasonic ethanol clean, show deep grooves — the equilibrium of grain-boundary energy against surface energy at the soak temperature — and grain boundaries that frequently traverse the full sheet thickness, a step toward columnar through-thickness grain structures<sup>21</sup> (Fig. 14). The best-correlated specimen, Ni4N5\_081 (annealed at a nominal 1300 °C for 20 h; logged soak  $\sim 1301$  °C), is documented across length scales — optical grain map with in-image grain-size measurements and high-magnification SEM grain-boundary detail (Fig. 15). The full multi-gigabyte raster/EBSD archive (including .oim, .osc, .ang, and DREAM.3D volumes) remains on the lab's institutional share and is inventoried per file in the characterization catalogs distributed with the data. Grain size is quantified with: number of grains analyzed, equivalent-circle-diameter or intercept metric, median + IQR (or mean  $\pm$  SD), segmentation/cleanup criteria, minimum-grain-area threshold, and whether twins are merged or separated, with the specimen/run (not the individual grain) treated as the experimental unit<sup>1,3,22</sup>. Where as-received baselines are unavailable for a batch, results are framed as final-state comparisons across archived anneals rather than as fully parameterized grain-growth kinetics.

Controls that further isolate the instrument's behavior — open-loop vs. closed-loop on the same target cycle, vacuum vs. inert-gas backfill mode, directly-coupled metal vs. susceptor-mediated ceramic heating, and multi-hour long-soak stability — are partially represented in the archive and are natural extensions for a prospective study.



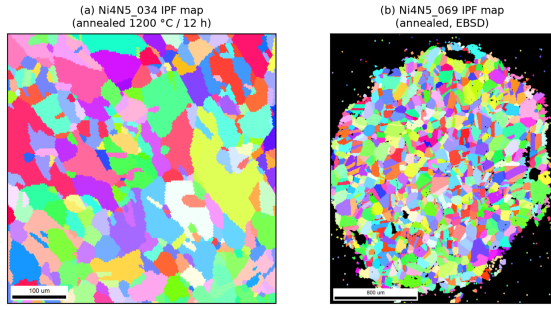


FIG. 12. EBSD inverse-pole-figure (IPF) orientation maps of two annealed Ni4N5 specimens, rendered from the archived EBSD scans. (a) Ni4N5.034 (annealed at 1200 °C / 12 h); (b) Ni4N5.069. Each indexed color is a crystallographic orientation; the maps resolve hundreds of grains for grain-size and texture analysis.

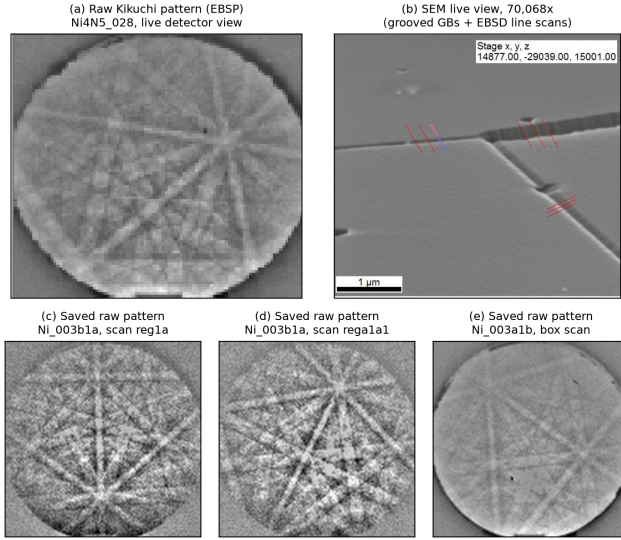


FIG. 13. Raw (as-collected) EBSD acquisitions from furnace-annealed nickel, with no metallographic preparation. (a) Raw Kikuchi pattern (electron backscatter pattern) of specimen Ni4N5.028 as seen live on the detector at  $8 \times 8$  binning (30 ms exposure), with Kikuchi bands and zone axes crisply resolved; (b) the corresponding SEM live view at 70,068 $\times$ , where thermal grooving alone delineates the grain boundaries crossed by the EBSD line scans (red marks; 1  $\mu$ m scale bar). (c-e) Representative per-scan-point raw patterns (115 $\times$ 115 px at  $8 \times 8$  binning) saved automatically during EBSD mapping of the early-campaign specimens Ni\_003b1a and Ni\_003a1b; more than 100,000 such patterns are archived. The pattern quality obtainable straight from the furnace reflects the low residual surface damage of the annealed condition.

*a. Specimen  $\leftrightarrow$  thermal-history linkage.* Each specimen used in the thermal-performance and microstructural validation is tied to its exact run — run ID, nominal and machine-parsed logged soak, duration, gas-flow state, and archived SEM/optical/EBSD file counts — in the specimen-run cross-reference table

distributed with the supplementary material, which also hyperlinks every entry to its raw run log and image data in the archive. The complete 35-specimen cross-reference includes characterized specimens whose run logs are not in the parsed set.

#### A. Extension to ceramic charges: YSZ

The same control, vacuum, and sensing stack extends to charges that do not couple to the RF field at all, demonstrated with grain growth in yttria-stabilized zirconia (YSZ). Ceramic grain growth requires substantially higher soak temperatures ( $\gtrsim 1700$  °C, and usefully 2000–2500 °C) than the Ni anneals — beyond the service range of the alumina spacers (1787 °C rating) and too hot for the graphite lid/sapphire-window geometry of Fig. 4 — so only the charge stack inside the quartz tube is exchanged (Fig. 17(a)). The YSZ specimen is sandwiched between two tantalum blocks (25.5 mm diameter) that replace graphite as the susceptor at these temperatures; the pair seats in a ceramic crucible (MgO shown; boron nitride was also used, see below) on an alumina support rod that raises the stack to coil height inside a fresh  $\sim 35$  mm-ID quartz tube. The generator, vacuum path, gas handling, pyrometer, and LabVIEW control are untouched — the swap is a specimen-stack change, not a furnace change. For these high-temperature runs the chamber was first evacuated and then continuously filled with argon while the turbopump was running, which helped avoid oxidation during heating.

*a. Tantalum-susceptor heat curve.* Figure 16 quantifies the temperature range this stack opens up, from an archived lab workbook. With a single tantalum block the pyrometer reads  $\sim 1230$  °C at a 10 % power command, rising monotonically to  $\sim 2020$  °C at 31 %; stacking two blocks shifts the whole curve upward ( $\sim 1790$  °C at 10 %), sustains a  $\sim 2000$  °C hold for over an hour at a constant 17 % command (Fig. 16(b)), and reaches 2344 °C at 33 % before that ramp was terminated by the chamber pressure interlock — i.e., the practical ceiling encountered so far is set by outgassing at temperature, not by available RF power. Since the dual-wavelength pyrometer reads to 2500 °C, the sensing chain covers the full demonstrated range.

*b. High-temperature material compatibility.* Operating between 2000 and 2500 °C makes chemical compatibility between the specimen, susceptor, and crucible the design constraint, and the archived configuration survey maps it empirically. Three stack variants were run, each seated in a boron nitride crucible resting on the alumina support rod: a tantalum susceptor block placed on top of the YSZ specimen at 2500 °C for 8 h (successful grain growth, but Ta vapor deposited on the YSZ); the YSZ specimen set into a graphite susceptor block at 2000 °C for 168 h (the YSZ reacted with the graphite and was consumed); and a Ta/YSZ/Ta double-block sandwich at 2130 °C for

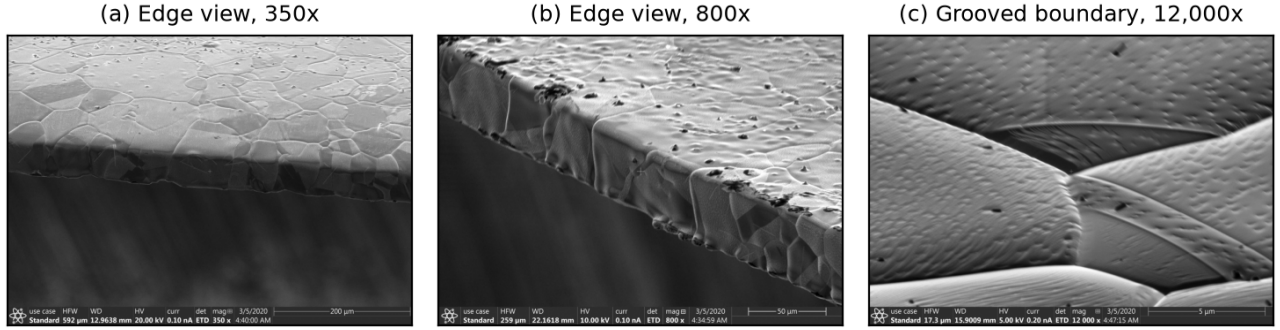


FIG. 14. Grain-boundary thermal grooving on specimen Ni4N5\_053 after a 1200 °C / 12 h anneal, imaged edge-on in the as-annealed condition (ultrasonic ethanol clean only; no polishing). (a) Edge view at 350 $\times$  (200  $\mu$ m scale bar): several grain boundaries traverse the full sheet thickness. (b) Edge and top surface at 800 $\times$  (50  $\mu$ m scale bar), with grooves outlining the surface grain structure. (c) A single grooved boundary at 12,000 $\times$  (5  $\mu$ m scale bar): the deep trench develops as grain-boundary energy equilibrates against surface energy at the soak temperature.

Ni4N5\_081 microstructure after its 1300 °C / 20 h anneal

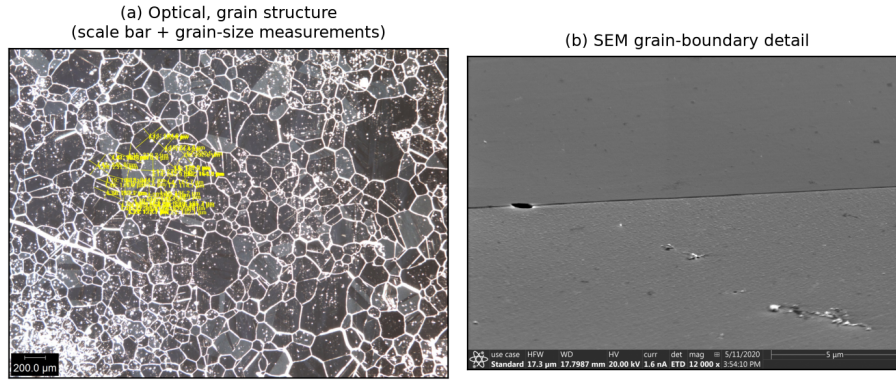


FIG. 15. Microstructure of specimen Ni4N5\_081 after its logged anneal (nominal 1300 °C / 20 h, gas flow), correlating the characterization archive with a single logged thermal history. (a) Optical micrograph (grain structure, 200  $\mu$ m scale bar, with in-image grain-size measurements); (b) high-magnification SEM grain-boundary detail.

80.4 h (a reaction shell formed around the tantalum, attributed to Ta–BN contact). The observed pairwise incompatibilities — alumina–graphite eutectic near 1800 °C; carbon consuming YSZ at 2000 °C; YSZ–Ta interdiffusion and Ta deposition at 2500 °C; Ta–BN and Ta–MgO surface degradation; Ta carburizing to (still-functional) gold-colored TaC; and Ta gettering oxygen from any source — are recorded with photographs in the archived survey. Two practical rules emerged: shorter, hotter anneals stay clean (a  $\sim$ 90  $\mu$ m clean-grained YSZ specimen was produced in 45 min in the double-Ta configuration, while runs past  $\sim$ 3 h contaminated the specimen), and above  $\sim$ 2200 °C a carbon susceptor that does not touch the YSZ (e.g. graphite contained in a BN crucible) is the indicated configuration. Archived optical micrographs (Fig. 17(b,c)) show the resulting equiaxed YSZ grain structures, including a specimen after a 1700 °C / 10 h anneal. Unlike the nickel anneals, most of the YSZ survey anneals were not logged under

the run-numbered log series, so they are reported from the archived survey record (heat-curve workbook, photographs, and notes) rather than from parsed run logs.

## B. Limitations of the retrospective validation

The validation dataset is retrospective rather than prospectively designed, so the archived runs are not treated as a balanced factorial experiment and comparisons are restricted to internally consistent subsets. Because induction coupling, radiative view factors, and pyrometer line-of-sight depend on specimen/crucible geometry and optical alignment, the power-command/temperature calibration is reported only for a fixed archived configuration and is not a universal transfer function. Reported temperatures are closed-loop ratio-pyrometer readings in a fixed

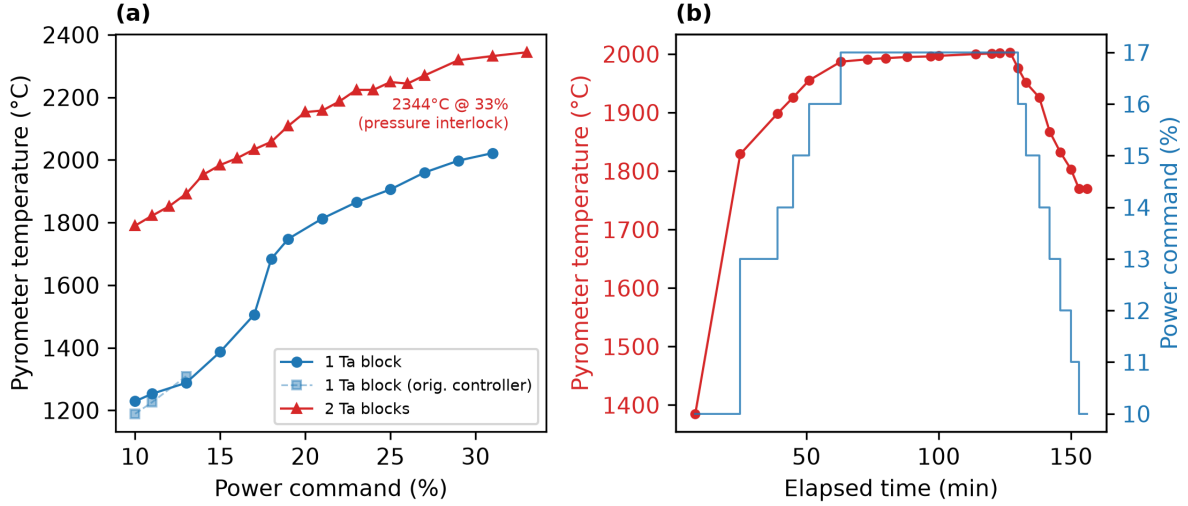


FIG. 16. Measured heat curves of the tantalum-susceptor (ceramic) stack. (a) Pyrometer temperature vs. generator power command for one and two stacked tantalum susceptor blocks (the faint series repeats the single-block measurement on the original power controller); the two-block ramp reached 2344 °C at a 33% command before the chamber pressure interlock ended the test. (b) Timed two-block ramp/soak: a constant 17% command holds ~2000 °C ( $2000 \pm 3$  °C over the final hour) before a stepped ramp-down.

optical geometry; absolute specimen temperature may still depend on emissivity and surface condition not fully reconstructable for every archived run<sup>16,17,23,24</sup>. Run-to-run repeatability is evaluated only within the most consistent cohort; most other temperature/time conditions are singletons, so inference is limited to descriptive summaries. Microstructural comparisons are reported only for specimens whose image files link unambiguously to a logged thermal history; ambiguously labeled images are excluded rather than inferred.

## VI. CONCLUSIONS

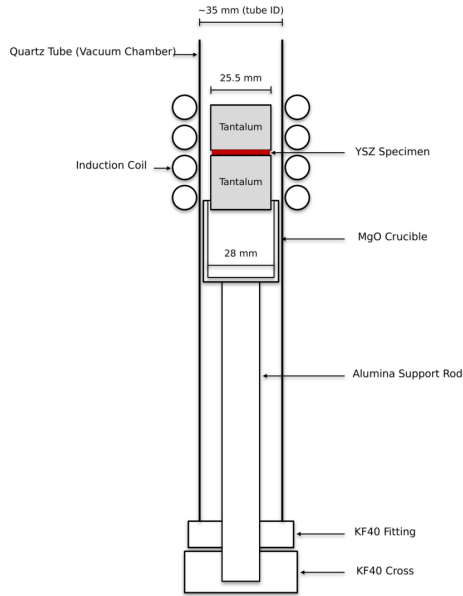
A bare commercial RF induction generator can be converted into a computer-controlled, vacuum-integrated, pyrometer-feedback annealing system with a documented, generator-agnostic modernization layer: LabVIEW/DAQ analog power control, dual-wavelength optical temperature feedback, a high-vacuum quartz-tube chamber, and an in-house machined graphite crucible/susceptor. Ported from a ~1970 LEPEL prototype to a current CEIA solid-state generator, the same stack demonstrated an operational power-temperature calibration ( $R^2 = 0.991$  over 1200–1400 °C in a fixed geometry), run-to-run soak repeatability of  $1201.2 \pm 1.3$  °C across eight nominally identical Ni4N5 anneals, and stable closed-loop soaks up to 40 h at 1325 °C, with annealed-specimen microstructures (SEM/EBSD/optical) cross-referenced to their logged thermal histories. Exchanging the graphite crucible stack for a tantalum-susceptor variant extends the same furnace to non-coupling ceramic charges —

sustaining 2000 °C holds and reaching 2344 °C before the chamber pressure interlock — demonstrated with YSZ grain growth and an empirical map of specimen/susceptor/crucible chemical compatibility at those temperatures. Because the layer requires only a monotonic analog power-control input from the generator, the design is directly reusable by other laboratories at a small fraction of the cost of turn-key vacuum induction furnaces; complete design files, the costed bill of materials, control software, and the full run-log and characterization archives are openly available.

## SUPPLEMENTARY MATERIAL

The development repository (<https://github.com/vertical-cloud-lab/custom-induction-furnace>; archival snapshot [10.5281/zenodo.20878017](https://zenodo.org/record/20878017)) contains the complete supplementary material: the LabVIEW control VIs and MATLAB utilities, the system schematics and coil drawing, the standard operating procedure, the itemized bill of materials and design-file inventory, the 100+ raw and machine-parsed anneal logs, the specimen-run cross-reference table (Table S1, with per-entry hyperlinks to the raw run logs and image data), the SEM/EBSD and optical characterization archives with per-file catalogs, the tantalum-susceptor heat-curve workbook and high-temperature configuration survey, and the scripts that reproduce every figure and metric in this paper.

(a)



(b)



(c)

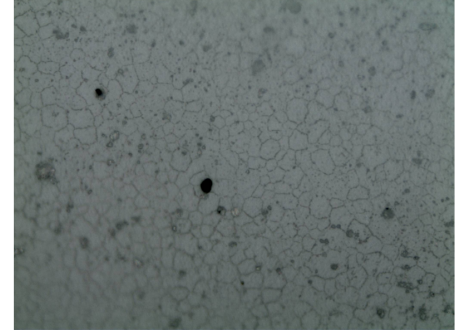


FIG. 17. High-temperature ceramic (YSZ) configuration and resulting microstructure. (a) Charge-stack schematic: inside a fresh  $\sim 35$  mm-ID quartz tube, the YSZ specimen (red) is sandwiched between two 25.5 mm tantalum susceptor blocks, seated in a 28 mm MgO crucible on an alumina support rod that raises the stack to coil height; the coil, vacuum, pyrometer, and control paths are unchanged from the metal configuration. (b) Optical micrograph of YSZ after a  $1700^\circ\text{C}$  / 10 h anneal. (c) Optical micrograph of an induction-annealed YSZ specimen, showing the equiaxed grain structure.

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## AUTHOR DECLARATIONS

### Conflict of Interest

The authors have no conflicts to disclose. **[TODO: confirm with all authors.]**

### Ethics Approval

This work did not involve human participants, human data, or animal subjects; ethics approval is not applicable.

## Author Contributions

**[TODO: confirm CRediT roles]** — provisional:  
**Sterling G. Baird:** Conceptualization, Software, Investigation, Writing – original draft.  
**Ryan Weber:** Investigation, Methodology, Data curation, Visualization.  
**Christopher Nyborg:** Investigation, Methodology, Data curation, Visualization.  
**Ronnie Guymon:** Visualization, Writing – review & editing.  
**Gage Erickson:** Visualization, Writing – review & editing.  
**Oliver Johnson:** Supervision, Project administration.

## DATA AVAILABILITY

The data that support the findings of this study are openly available in the development repository `vertical-cloud-lab/custom-induction-furnace` (<https://github.com/vertical-cloud-lab/custom-induction-furnace>); an archival snapshot (design files, run logs, characterization data, and extracted context) is deposited at Zenodo, <https://doi.org/10.5281/zenodo.20878017>.



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